SAKIB BIN FARUQUE RUSHO

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Career Objective

AI & Software Engineer specializing in Computer Vision, Deep Learning, and mobile ML deployment. Experienced in integrating edge pose estimation models into production Android applications, training neural architectures for object detection, and implementing generative AI frameworks. Seeking an AI/ML Software Engineer role to build production-grade, model-driven applications.

Education

University of Rajshahi 2020 – 2025
B.Sc in Computer Science & Engineering CGPA: **3.52/4.00**

Experience

Software Engineer (Android & AI Integration) Jan 2026 – Present
MyMedicalHub International

- Engineer native Android applications (**EMMA Health**, **Atlas Athlete**, **AI Guided Therapy**) utilizing **Kotlin**, **Jetpack Compose**, and **MVVM** / **Clean Architecture**.
- Implement computer vision video processing logic featuring frame-level pose analysis and automatic presence detection to control session triggers.
- Build automated session feedback loops, dynamic exercise summary visualizers, and pre-assessment compliance checks.
- Maintain production CI/CD workflows, strict Git versioning, and collaborative Agile development routines.

Junior Flutter Developer Mar – May 2025
ScaleBridger

- Developed production-grade Flutter applications integrated with Firebase backend services using Clean Architecture and Cubit/Bloc.

Thesis

Integrating CBAM and CLAHE in RetinaNet for Enhanced Remote Sensing Object Detection
Developed a novel RetinaNet-based framework combining attention mechanisms and contrast enhancement for aerial imagery. Achieved **93.47% mAP@0.5** on the RSOD dataset.

Deep Learning & AI Projects

- **Face Verification using Deep Learning**
  Trained and evaluated Siamese neural networks using **BCE**, **Contrastive**, and **Triplet Loss** on WIDER FACE and Anime datasets using PyTorch.
- **GAN Variants for Face Generation**
  Implemented **DCGAN**, **Conditional GAN**, and **CycleGAN** for synthetic face generation, attribute transfer, and image-to-image translation in TensorFlow.
- **Crowd Counting & Density Segmentation**
  Trained **U-Net** and **MCNN** architectures on ShanghaiTech Part_A for crowd density estimation, evaluated via MAE and RMSE metrics.
- **Knowledge Distillation and Transfer Learning**
  Implemented soft-label batch-wise distillation between **ResNet50**, **VGG16**, and compact student CNNs to optimize edge model inference speed with minimal accuracy loss.

Technical Skills

Programming: Python, Kotlin, C++, Java, Dart, SQL
Frameworks & Libraries: PyTorch, TensorFlow, Keras, OpenCV, Jetpack Compose, Flutter
Computer Vision & AI: Frame-Level Pose Estimation, YOLO, RetinaNet, U-Net, MCNN, Siamese Networks, CLAHE, CBAM
Generative AI & Training: DCGAN, cGAN, CycleGAN, VAE, Transfer Learning, Knowledge Distillation
ML & Math Tools: NumPy, Pandas, Matplotlib, Grad-CAM, Integrated Gradients, PCA
Developer Tools: Git, GitHub, Azure DevOps, Google Colab, Jupyter, LaTeX

Achievements

- Ranked 7th in KUET Datathon (Data Science Competition).
- 2nd Runner Up in Flutter Stack, Learnathon 3.0.
- Completed **LEARNATHON 3.0** intensive mobile application training program (4 months).